

Kush Garg

Software Engineer II | Backend & Distributed Systems | SWIFT ISO 20022

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SUMMARY

Software Engineer II with 2+ years building scalable payment platforms at StoneX Group. Own end-to-end delivery of the SWIFT ISO 20022 FAAS pipeline and co-shipped XPAY, enabling \$600M in operational savings. Skilled in Java, Spring Boot, Kafka, and microservices with applied AI experience (LangGraph, multi-agent orchestration). ICPC Regionals qualifier, Google Kickstart Global Rank 1596.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, C++

Frameworks & Databases: Spring Boot, Spring Security, Spring WebSocket, Spring Cloud Gateway, GraphQL, FastAPI, Node.js, React, Next.js, MongoDB, PostgreSQL, Redis, MySQL

AI & Infrastructure: LangChain, LangGraph, scikit-learn, Docker, Kafka, Nginx, Azure DevOps, GitHub Actions, CI/CD

Domain: SWIFT ISO 20022, pacs.008, pacs.009, pain.001, CAMT.053, CAMT.054, MT900, MT910, MT942, Microservices, System Design, API Design, Agile/Scrum

PROFESSIONAL EXPERIENCE

Software Engineer II - StoneX Group, Pune, IN

Aug 2024 - Present

- Own end-to-end delivery of the Funding as a Service (FAAS) pipeline, a centralized payment platform serving all StoneX entities with full ACK/NACK lifecycle management.
- Built ISO 20022 message generation layer (pacs.008, pacs.009, pain.001) ingesting TMSJson payloads from Treasury Management Systems and delivering to the SWIFT network.
- Engineered SWIFT message processing layer supporting 7+ message types across legacy MT (MT900, MT910, MT942) and modern MX (CAMT.053, CAMT.054) formats.
- Co-drove XPAY to production (Feb 2025), processing cross-border payments from the CONNECT app and broker initiations, enabling \$600M in operational savings.
- Proposed AI cost-reduction architecture using GraphDB + VectorDB + MCP with CI/CD automation to eliminate redundant LLM token consumption in code-data sync workflows.
- Led team of 5 to 2nd place at StoneX India Hackathon (Nov 2025), building an AI ops tool with multilingual chatbot and interactive dashboard for operations users.

Software Engineering Intern - StoneX Group, Pune, IN

Jan 2024 - Jul 2024

- Core contributor to XPAY: built Spring Boot REST APIs, transaction logic, compliance hooks, and audit trails forming the foundation of its production payment architecture.
- Delivered production CI/CD pipelines on Azure DevOps with DevOps and InfoSec teams, meeting compliance standards for go-live.

PROJECTS

NeoTrade - Rules-First Trading Engine for Indian Markets

Python, FastAPI, LangGraph, MongoDB, Redis, React 19 | github.com/kushgarg132/NeoTrade (Live: neotrade-trading.vercel.app)

- Built a strategy engine for Indian equities and F&O emitting fully-sized trade intents (entry, stop, target, rule codes, composite score); intraday trades auto-execute, long-term proposals route to an approval inbox with 3-day expiry.
- Capped AI contribution to trade conviction at 30%, enforced in the type system so the LLM cannot rescue a trade the rules reject; added risk-aware position sizing, backtesting with Indian transaction costs, and Zerodha Kite Connect integration.
- LangGraph research and chat agents via a self-hosted LLM gateway; 115 pytest test files; push-to-deploy via GitHub Actions to Docker Compose on Oracle Cloud and Vercel.

Betrix - AI-Powered Multiplayer Poker Platform

Java 21, Spring Boot 3.5, MongoDB, GraphQL, WebSocket, Gemini AI, React | github.com/kushgarg132/Betrix (Live: betrix-b3c24.web.app)

- Built full-stack Texas Hold'em platform with Gemini AI bot opponents (random-play fallback under rate limits or outages) and GraphQL subscriptions over WebSocket for real-time game state sync; engineered stateful poker logic: hand evaluation, pot calculation, side pots, dealer rotation.
- Secured with Spring Security JWT plus guest login; added admin game replay from an event log; deployed via Docker Compose on Oracle Cloud behind Nginx + Let's Encrypt, frontend on Firebase Hosting, push-to-deploy on both.

Chess Platform - Distributed Microservices

Java 21, Spring Boot 3, Spring Cloud Gateway, PostgreSQL, Redis, WebSocket, Next.js | github.com/kushgarg132/Chess

- Designed 7 Spring Boot services behind Spring Cloud Gateway with JWT auth: WebSocket game state, ELO-bracketed matchmaking on a Redis queue, and a pooled Stockfish engine service for AI play and move analysis.

RoutineOS - Personal Discipline & Routine Tracker

Java 21, Spring Boot 3, PostgreSQL, Flyway, React, TypeScript (Live: frontend-seven-pied-17.vercel.app)

- Discipline tracker with deterministic scoring, streaks, and heatmaps; timezone-aware sweeps and FCM push notifications.

EDUCATION

B.Tech in Computer Science (AI and ML Specialization) - Symbiosis Institute of Technology, Pune

Aug 2020 - Sep 2024

AWARDS & CERTIFICATIONS

Google Kickstart 2021 Round B - Global Rank 1596 | ICPC Asia Kanpur Regionals 2021-22 - qualifier (Team Volatile Voids)